

Manufacturing and supply chains

- **Preventing machine failures:** Predictive maintenance keeps equipment running smoothly by identifying issues before they cause downtime.
- **Optimising logistics:** AI finds the most efficient delivery routes, reducing costs and delays.
- **Improving quality control:** AI-powered inspection systems catch defects faster and more accurately than humans.

Education and research

- **AI tutors that adapt to students:** Personalised learning tools adjust to each student's progress.
 - **Making academic research easier:** AI speeds up literature reviews by summarising key insights from thousands of papers.
 - **Breaking language barriers:** AI-powered translation tools make learning materials accessible to a global audience.
- From finance to education, open source LLMs are proving to be game-changers—offering customised, scalable solutions without the constraints of proprietary systems.

Security in open source LLMs

Security is always a major concern when implementing AI, and open source models come with their own set of advantages and challenges. The good news? They offer far more transparency and control than closed source alternatives.

Full transparency and security audits: Unlike proprietary models, open source LLMs allow developers to inspect every line of code and review training data. This means security flaws can be spotted and fixed faster by the global developer community.

On-premise deployment for maximum privacy: Companies worried about data security can choose to host these models on their own infrastructure, keeping sensitive information completely in-house. This eliminates risks tied to third-party cloud providers.

Customisable security layers: Because the code is open, businesses can add their own encryption, authentication,

and access controls, ensuring their AI system meets specific security requirements.

A global community for faster fixes: Unlike closed models where users depend on a single company for updates, open source AI benefits from a worldwide network of developers who actively monitor and improve security.

So, while security challenges exist, open source models provide something invaluable—the ability to take control rather than relying on a black-box solution. Until recently, training and fine-tuning LLMs required enormous computational resources, limiting AI development to well-funded organisations. Open source LLMs are breaking down these barriers by allowing anyone to experiment with and deploy AI models tailored to their needs.

For example:

- **Falcon** delivers state-of-the-art performance with efficiency, making it a great option for enterprises looking for cost-effective AI.
- **Llama 3 (Large Language Model Meta AI 3)**, Meta's latest release, balances power and accessibility, enabling developers to integrate advanced NLP into their applications.
- **BLOOM (BigScience Large Open-science Open-access Multilingual Language Model)**, created by Hugging Face, is a multilingual powerhouse designed to support diverse global applications. It has 176 billion parameters for building transformer-based autoregressive LLMs.
- **BERT (Bidirectional Encoder Representations from Transformers)** from Google, one of the earliest and most influential open source NLP models, continues to be widely used for search, chatbots, and more.

These models are not just alternatives to proprietary solutions—they're often just as capable, if not better in some cases, depending on the use case.

How open source LLMs handle AI hallucinations

AI hallucination refers to instances where a language model generates content that is factually incorrect, logically

Table 1: Comparison of the features of different open source LLMs

Model	Developer	Strengths	Best use cases	Licence
Falcon	TII	High efficiency, strong performance	Enterprise applications, chat-bots	Apache 2.0
Llama 3	Meta	Powerful, optimised for fine-tuning	AI assistants, research, apps	Custom (free for research and commercial use with conditions)
BLOOM	Hugging Face	Multilingual, diverse dataset	Global applications, translation	Apache 2.0
Mistral 7B	Mistral AI	Compact yet powerful, strong reasoning	Lightweight AI applications, code generation	Apache 2.0
BERT	Google	Optimised for NLP tasks, widely used	Search engines, sentiment analysis	Apache 2.0